



Dissertation on
“Real-Time Inappropriate Speech Detection”

Submitted in partial fulfilment of the requirements for the award of the degree
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In
Computer Science & Engineering (AI&ML)
UE22AM441A – Capstone Project Phase - 3

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CERTIFICATE

This is to certify that the dissertation entitled

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In partial fulfilment for the completion of seventh semester Capstone Project Phase - 3 (UE22AM441A) in the Program of Study -Bachelor of Technology in Computer Science and Engineering (AI&ML) under rules and regulations of PES University, Bengaluru during the period August – Dec 2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 7th semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Capstone Project Phase - 3 entitled “**Real-Time Inappropriate Speech Detection**” has been carried out by us under the guidance of **Dr Jayashree R, Professor & Chairperson Department of CSE(AI&ML)** and submitted in partial fulfilment of the course requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (AI&ML)** of **PES University, Bengaluru** during the academic semester August – Dec, 2025. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

There have been serious challenges to keep online platforms safe and respectful due to unmoderated content because of increasing incidents of explicit speech. The importance of real-time inappropriate speech detection lies in its ability to censor profanity which makes online platforms safe for all demographics.

The objective of this work is to clearly define our problem statement and its scope, display our learnings from our literature survey, highlighting research and technological gaps, define our objectives and conclude our findings while planning for future work. Our work focuses on the problem of detecting inappropriate speech in real-time, which has not been explored yet.

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3.16 Research Paper – 16

“AudioVSR: Enhancing Video Speech Recognition with Audio Data,”

3.17 Research Paper – 17

“Prediction of User Emotion and Dialogue Success Using Audio Spectrograms and Convolutional Neural Networks,”

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Chapter 1

INTRODUCTION

1.1 The increasing need for Content moderation in online Platforms

Due to recent increase in popularity of Live Streaming platforms, there exists a need for establishing a safe environment for all age demographics by moderating use of profanity that may or may not be intentional. Prediction of occurrence of inappropriate speech has not yet been explored in depth as compared to Traditional moderation techniques which rely on text or audio solely and are not used for Real-Time detection.

1.2 The Approach of Multimodal Models in Predicting Profanity

Multimodal models can offer a nuanced approach by simultaneous analysis of audio, video and text to overcome the limitations of unimodal systems. The model can extract temporal sequences and spatial features. By incorporating the use of deep learning architectures like Convolutional Neural Network and Recurrent Neural Networks, we can increase efficiency.

1.3 Scope of Proposed Framework

We are proposing the use of a multimodal deep learning framework for detecting inappropriate language in real-time systems. With the help of a combination of audio and visual characteristics, and cross - modal attention mechanisms, the model will become better at forecasting inappropriate content by concentrating on the important features in these various modalities.

The long-term objective is to come up with a context - aware system, which may allow platforms to create a safe environment.

Chapter 2

PROBLEM STATEMENT

2.1 The Challenges of Moderating Live Streaming Platforms

It is difficult to moderate content on live streaming platforms because everything is unscripted. Live streams require continuous monitoring because of its unpredictable nature, which makes it harder to catch inappropriate content such as curse words. Also, real-time live streams are not just about one kind of data type but include audio, video, and text all at once. This variety makes moderating even more challenging to do unless you have a sophisticated model capable of analysing all inputs simultaneously.

2.2 The Problem of Predicting Inappropriate Language

Foul words need not always be spoken out but also can also depend on the body language and facial expression. Along with this sarcasm and irony can also be difficult to predict only on either spoken words or body language.

2.3 Why We Need Multimodal Analysis

A complete picture is difficult to get with traditional single-modal models. When audio, video, and text are all analysed together as a multimodal approach, we can improve the accuracy of identifying inappropriate language and behaviour.

It is easier to figure out the full context of what is happening when we combine speech with visual data. Also, by using many modalities of data, the system can get a better understanding and become more reliable. Similarly, only multimodal systems can notice and predict even the most subtle forms of curse words.

CHAPTER 3

LITERATURE SURVEY

3.1 Research Paper – 1

3.1.1 Paper Title

I. Smirnov and A. Laushkina, "Multimodal prediction of profanity based on speech analysis," [1]

3.1.2 Summary

The key idea behind this paper was to develop a real-time profanity prediction model, using audio features, to identify swearing before the full word is spoken. This helps with the task of early censorship. They found that using audio features would be better than Automatic Speech Recognition (ASR) as it can often lag and is prone to errors in certain conditions. This provides possibilities to combine with facial expression analysis to build a multimodal model which gives better results. The audio features were extracted using methods like MFCC and Wav2Vec. For the speech recognition task, they used Whisper ASR for transcribing to speech, and also label timestamps showcasing profanity. It created 3 class labels – Silence, Normal speech, Profanity. The CommonVoice and CMU MOSEI dataset were used with deep learning models like, LSTM and Bidirectional LSTM with Attention mechanism, this model improves prediction by understanding the context compared to LSTM which only predicts the upcoming words in speech. The real-time processing was executed by a pipeline which the authors implemented by using Voice Activity Detection (VAD), which initiates the prediction process upon detecting active speech, and Sliding window processing to analyse chunks of continuous audio. The best model – LSTM + Wav2Vec + Attention, achieved a F1-score of 92%. The idea of incorporating facial analysis was proposed by the authors to build even more robust models.

Approach	F1 score (The main dataset)	F1 score (MELD dataset)	F1 score (Picked records)	Latency (sec)	Weights (mb)
LSTM (MFCC)	86.6	65.2	73.7	0.347*	51.53
LSTM (MFCC) + Attention	87.1	65.9	76.7	0.348	231.16
LSTM Wav2Vec + Attention	92.0	71.4	86.6	1.171	432.25**
LSTM (MFCC) + ASR + Attention	90.3	67.6	74.6	2.148	231.19

* - the average duration of the element is 0.34 sec

** - it is also necessary to use Wav2Vec feature extractor

Fig. 3.1.1 Comparison of different LSTM architectures

3.2 Research Paper – 2

3.2.1 Paper Title

A. Chaudhari, P. Davda, M. Dand and S. Dholay, "Profanity Detection and Removal in Videos using Machine Learning," [2]

3.2.2 Summary

The authors tackled the issue of automatically finding and getting rid of swear words in videos. As manually doing this task consumes time, they came up with a system using machine learning to handle it, aiming for real-time detection and censorship. They start by using an approach that gets the audio from video and then converts it into text. Then, using a pre-made list of profanities they check these words. On appearance of a swear word, two things were done: they mute that specific part of the audio track, and they also visually censor the speaker's mouth by pixelating their lips. To perform the lip pixelation they recognize the face within the video, locate the lips landmarks (with the help of HOG and SVM) and only then apply the blur effect to the lips when the swear word is pronounced. To determine the specific moments when the profanity takes place, they initially subdivide the video into 3-second segments and then reduce it to 1-second sections. This was found to be more accurate. To conduct the testing, they did not use a standard dataset, but collected their own videos with profanity in locations such as YouTube and Facebook. They tested it on 50 videos, which had 187 swear words distributed throughout it. The system actually did pretty well, managing to catch and block the profanity correctly about 82% of the time (specifically, they reported 82.35% accuracy). They noted that using both the sound muting and the lip blurring together really did the trick for getting rid of the unwanted content.

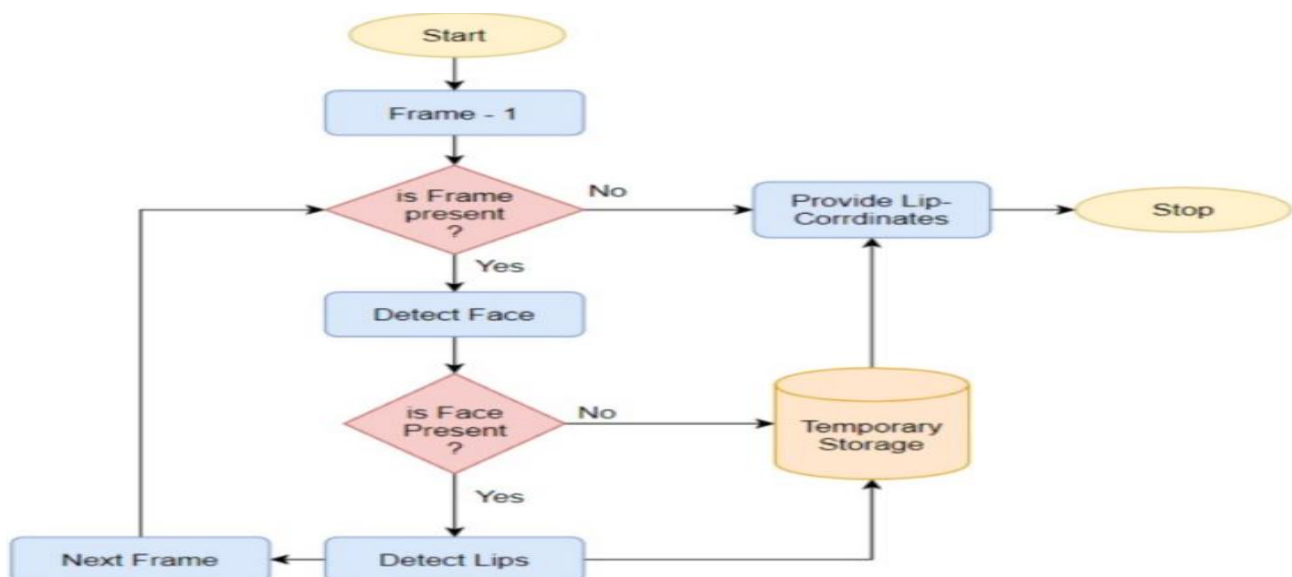


Fig. 3.2.1 Lip-Landmarks Detection methodology

3.3 Research Paper – 3

3.3.1 Paper Title

Rana, Aneri and Sonali Jha, “Emotion Based Hate Speech Detection using Multimodal Learning,” [3]

3.3.2 Summary

The authors of this paper created the first multimodal deep learning framework that combines audio (emotion detection) and text features to detect hate speech in videos. This is really hard to do because it needs processing of different types of data simultaneously. Their model does 2 things - the first one is text processing using BERT & ALBERT (Transformer models) trained on hate speech datasets from Twitter and Reddit, and the second thing is emotion detection in speech using a multi-task deep learning model trained on IEMOCAP dataset to detect valence (positive/negative emotion), arousal (intensity), and dominance (speaker control). They combined these features using a Multilayer Perceptron model for final classification. They also made their own Hate Speech Detection Video Dataset (HSDVD) with videos from Twitter & YouTube. This approach proposed by the authors is useful because it can help identify moments when a speaker might swear based on tone and intensity, which could enable real-time censoring. Their findings demonstrated that the multimodal learning, i.e., a combination of text and speech characteristics, worked better than the text-only models. This was showcased using the combination of BERT and Speech Emotion features, which produced the best results. One of the most intriguing results was that their model did not make many errors and falsely categorized sarcasm and neutral speech as hate speech. This flaw was seen in the text-only models which wrongly identified them as hate speech.

3.4 Research Paper – 4

3.4.1 Paper Title

Ba Wazir, A.S.; Karim, H.A.; Abdullah, M.H.L.; AlDahoul, N.; Mansor, S.; Fauzi, M.F.A.; See, J.; Naim, A.S., “Design and Implementation of Fast Spoken Foul Language Recognition with Different End-to-End Deep Neural Network Architectures,” [4]

3.4.2 Summary

The authors of this paper came up with lightweight deep learning models, i.e., CNNs and RNNs, to detect bad language in speech within a short amount of time, aiming to make real-time censorship achievable. The problem is that most of available models are extremely large and cannot be used live. They tested both CNNs trained on Mel-Spectrograms in order to extract visual patterns out of speech and RNNs with LSTM Cells that are capable of dealing with speech in real time. The data that the authors dealt with is the MMUTM foul language dataset that consists of 9 types of offensive words and regular speech of 117 different individuals under different circumstances. They modified the data by making the pitch different, altering frequencies, and introducing noise to make their system more robust, in other words, they increased the size of the dataset by about 9 times which made the system more effective in overcoming background noise. Their CNN model, with only 57K parameters, is likely the best part as it can actually be used in real-life scenarios as it is relatively lightweight. They did tests under varying noise conditions between clean to very noisy in order to establish the effectiveness of the models. Their CNN model had the better F1-score with 96.11% compared to the RNN model which scored 94.85%. They could keep up with massive pre-trained networks like AlexNet and ResNet50 with their smaller CNN model. In noisy cases, the efficiency was reduced by 10% but the data augmentation undertaken by the authors mitigated the effect. This piece is highly applicable since the LSTM methodology is able to detect swear words when they are being pronounced and not when they are over. This is precisely what is required in live video censoring.

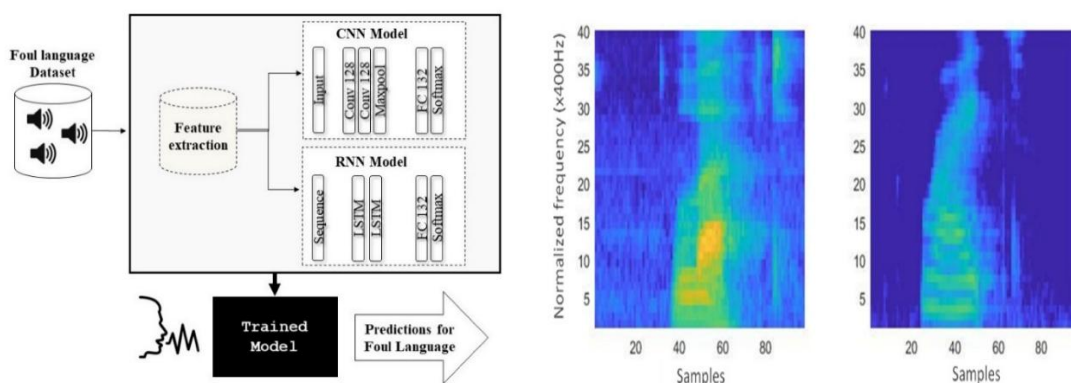


Fig. 3.4.1 Samples and Workflow of the Foul Language Detection model

3.5 Research Paper – 5

3.5.1 Paper Title

J. Pan, W. Fang, Z. Zhang, B. Chen, Z. Zhang and S. Wang, "Multimodal Emotion Recognition Based on Facial Expressions, Speech, and EEG," [5]

3.5.2 Summary

The authors developed this system known as Deep-Emotion which essentially involves a combination of facial expressions, speech and brain signals (EEG) to better understand the feelings of people. The remaining available systems utilize just one or two types of data, which is not as good. To analyse faces, the authors applied a GhostNet CNN model that better captures emotion features and generalizes less. For speech emotion, the authors created this lightweight network named LFCNN, which is not very large and can, therefore, be used in real-time. They also incorporated EEG analysis with this Tree-like LSTM model to keep an eye on the brain activity. The fact that they have added all this together, i.e., using a Decision-Level Fusion Method with optimized weights to unite all three predictions, is likely to be the highlight of this paper. Their findings were fairly strong, their face model achieved 98.27% accuracy, their speech model achieved 94.36%, and their EEG model also performed well. This demonstrated that mixing various signals is certainly more effective than utilizing a single signal. This research is relevant because the face and speech emotion analysis could help identify when someone's getting frustrated or angry right before they swear. Their fusion approach could definitely be adapted to mix speech and facial cues to predict swear words in videos, and since their speech model is small, it could actually work for real-time processing.

3.6 Research Paper – 6

3.6.1 Paper Title

H. M. Fayek, M. Lech and L. Cavedon, "Towards real-time Speech Emotion Recognition using deep neural networks," [6]

3.6.2 Summary

The authors developed a model for real time speech emotion recognition using a deep learning model to skip the steps of data preprocessing such a complex feature engineering due to the fact that deep learning models do their own feature analysis. They suggested using one second frames of raw speech spectrograms and also implemented techniques such regularization and data augmentation to combat overfitting as well as improve generalizability. The model achieved good accuracy on both the eNTERFACE and SAVEE datasets.

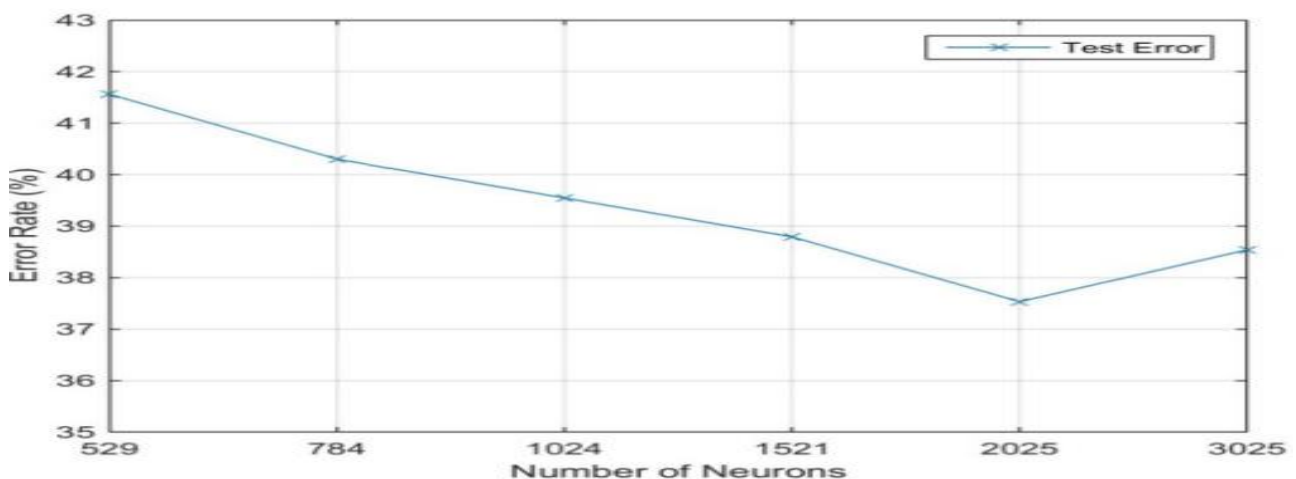


Fig. 3.6.1 Test Error Rates with varying neurons and fixed layers

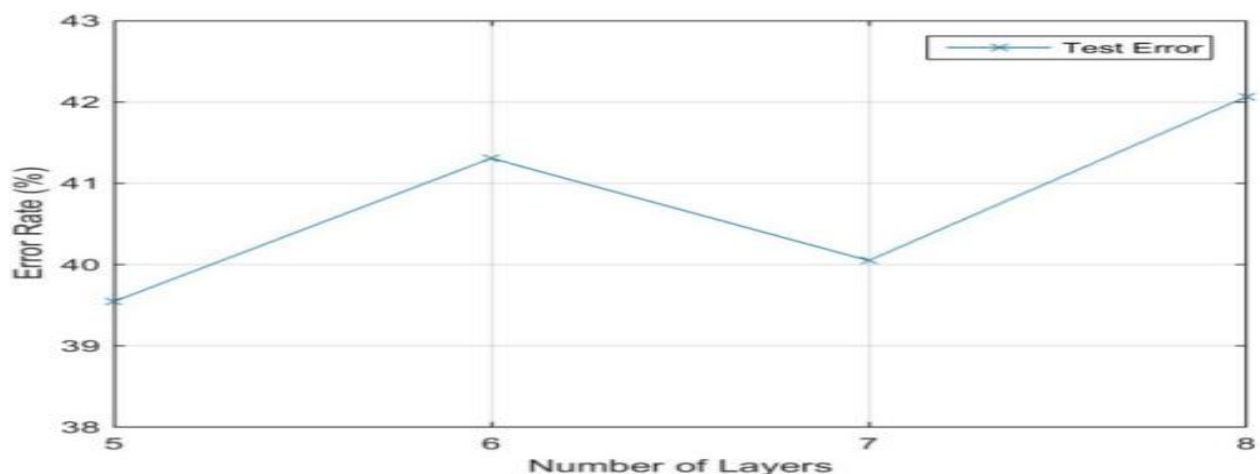


Fig. 3.6.2 Test Error Rates with varying layers and fixed neurons

3.7 Research Paper – 7

3.7.1 Paper Title

H. Meng, T. Yan, F. Yuan and H. Wei, "Speech Emotion Recognition From 3D Log-Mel Spectrograms With Deep Learning Network," [7]

3.7.2 Summary

The authors use a deep learning architecture for speech emotion recognition that consists of CNN for spatial feature extraction and a BiLSTM model with attention for temporal feature extraction. Coming to the loss function they used a hybrid loss function of softmax and cross entropy. The model showed an improvement of 4.58% over other models.

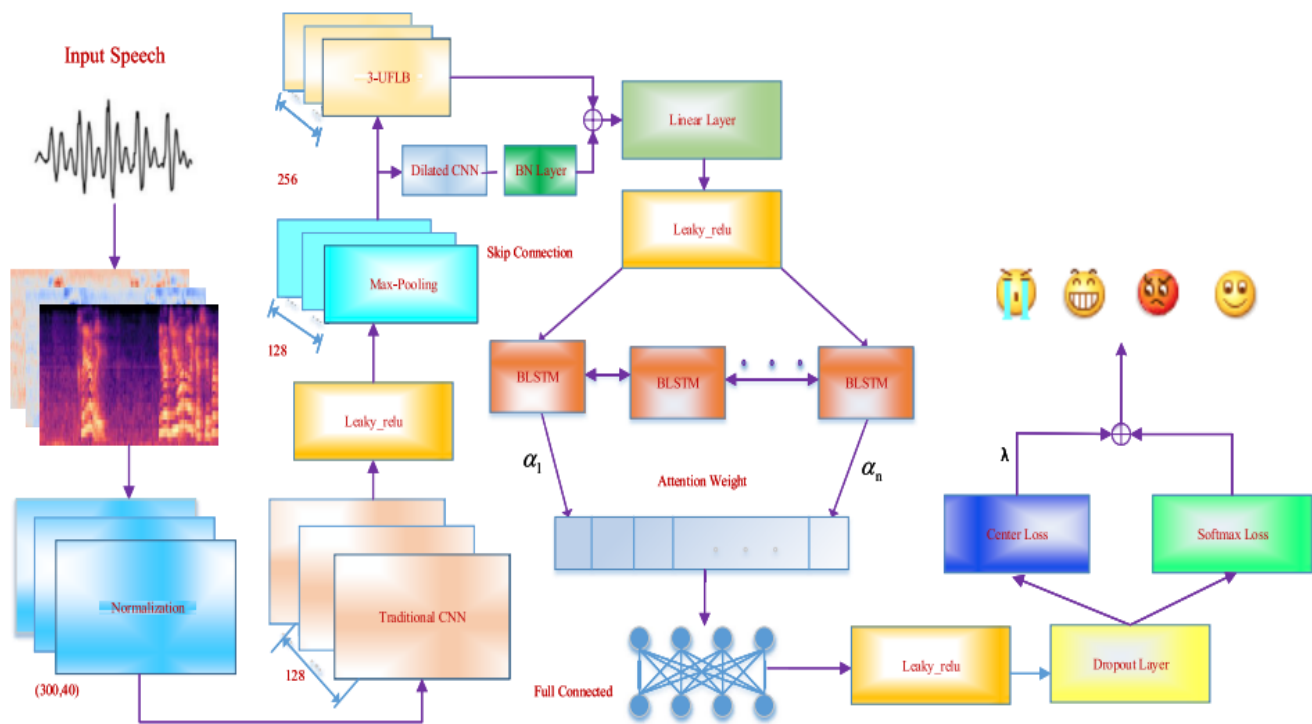


Fig. 3.7.1 Structure of dilated CNN and BiLSTM

3.8 Research Paper – 8

3.8.1 Paper Title

D. Duncan, G. Shine, and C. English, "Facial Emotion Recognition in Real Time," [8]

3.8.2 Summary

The authors used CNNs to perform real time facial emotion recognition in video feeds. They implemented a pipeline to capture live video, detect facial landmarks, feed it to the CNN and recognise emotions such as happiness, anger, sadness, etc. The authors designed this pipeline to improve as much speed and performance as possible. The final findings were that CNNs are quite capable at real time emotion recognition.

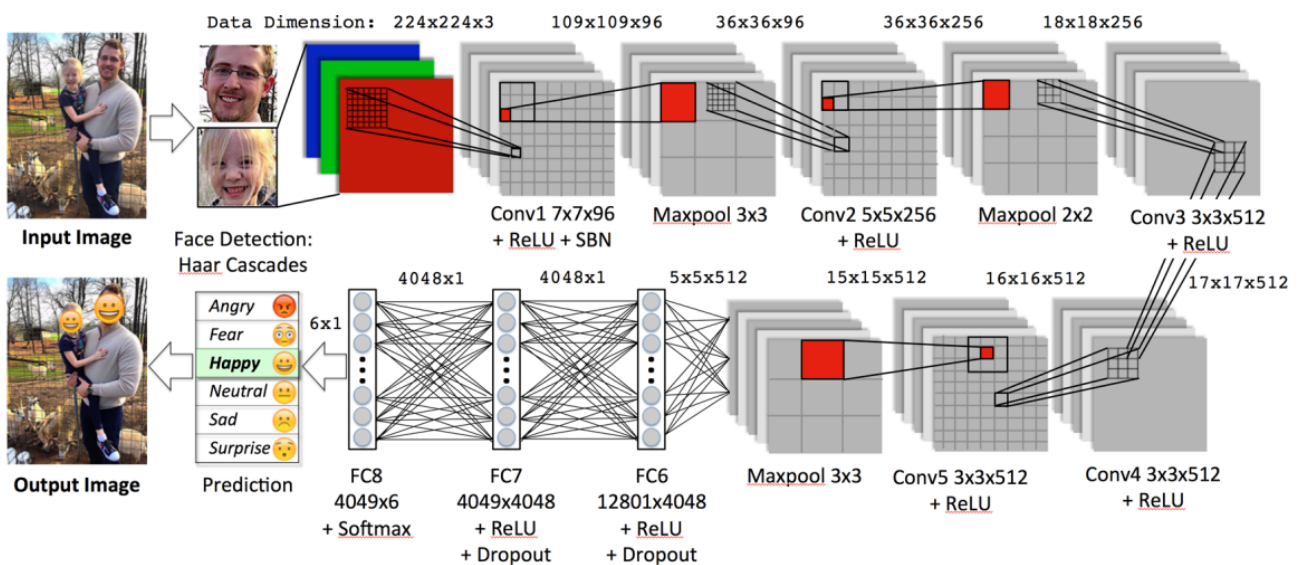


Fig. 3.8.1 CNN architecture with a Haar-Cascade detector



Fig. 3.8.2 Emotions detected by the CNN

3.9 Research Paper – 9

3.9.1 Paper Title

S. Dwijayanti, M. Iqbal and B. Y. Suprpto, "Real-Time Implementation of Face Recognition and Emotion Recognition in a Humanoid Robot Using a Convolutional Neural Network," [9]

3.9.2 Summary

The authors researched the possibilities of integrating facial emotion recognition in robots. They compared AlexNet, VGG-16 and a custom model for this task on data collected from human participants. The authors found that VGG-16 performed best on both the facial recognition and emotion recognition tasks with 100% and 73% accuracy respectively. The research concluded that VGG-16 is highly effective for facial emotion recognition tasks.

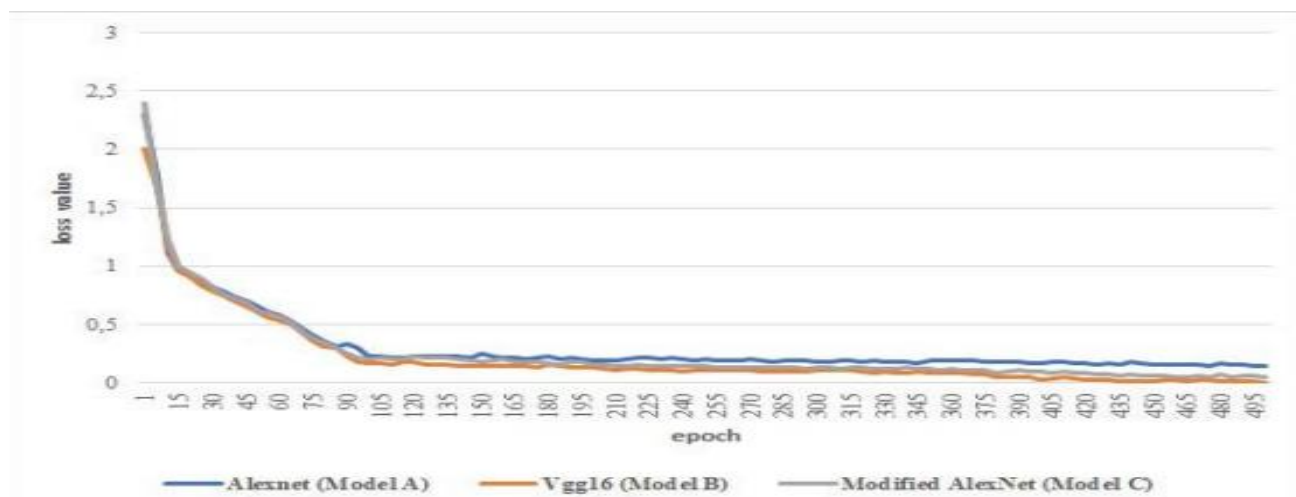


Fig. 3.9.1 Training Loss b/w models in Face Recognition

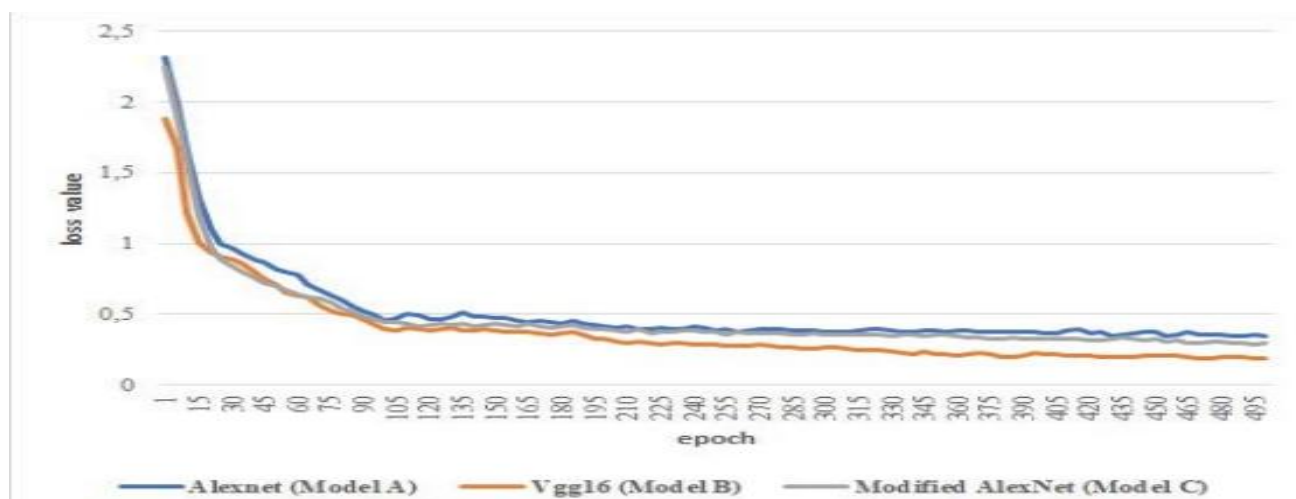


Fig. 3.9.2 Training Loss b/w models in Emotion Recognition

3.10 Research Paper – 10

3.10.1 Paper Title

Joan Serrà, Ilias Leontiadis, Dimitris Spathis, Gianluca Stringhini, Jeremy Blackburn, and Athena Vakali. 2017, “Class-based Prediction Errors to Detect Hate Speech with Out-of-vocabulary Words,” [10]

3.10.2 Summary

The main challenge addressed by this paper involves detecting hate speech that includes Out-of-Vocabulary (OOV) words, which consists of offensive words that are intentionally misspelled and new unique slurs. Most of the existing methods were basic involving n-gram count and word embeddings, and these models struggle with such ever-evolving language. The authors use prediction errors generated from class-specific models in order to predict these OOV words. The methodology put forward includes Class-specific language models, Error signal extraction, and Neural Network Classifier. In class-specific LMs, the training involved individual character-level models for each label, like hate speech or non-hate speech. These class-specific LMs train on data of their respective class labels and learn to predict the next character. For Error signal extraction, a prediction error, such as Cross-Entropy loss, from each class-specific LM is computed for a given input text which results in a set of error signals showing how well each model makes predictions. These signals are passed to the Neural Network Classifier that provides the likelihood of the input text belonging to each class. According to the authors, this approach helped improve the model’s generalisation. The test was conducted on a dataset of abusive tweets with higher percentage of OOV words. This test showed that the model outperformed the old methods by 4% - 11% during classification tasks. We find that this paper shows the importance in trying to stay updated with the evolving linguistic terms used by and influencing a great part of the online society in everyday life.

3.11 Research Paper – 11

3.11.1 Paper Title

H. Bhin, Y. Lim and J. Choi, "Multimodal Personality Prediction: A Real-Time Recognition System for Social Robots with Data Acquisition," [11]

3.11.2 Summary

This paper sees the author aim to predict human personalities in real-time with a goal to enhance human-robot interactions. After analysing personalities, they believe robots can alter their responses based on each individual. This analysis helps in creating more natural response and behaviour to different situations. The authors find this crucial for social robots to interact in the same manner as humans, such as showing empathy. This is much more effective as it is not as time consuming as traditional methods and is feasible for real-time applications, with multimodality enabling robustness for such operations. The multimodal system consists of Data Acquisition followed by Feature Extraction before Model Training and finally implementing Real-Time Predictions. The initial step starts with collection of data like audio and video cues from human interactions. This is followed by extracting the relevant information which could indicate personality traits. The model is then trained on these features to make predictions of 5 class labels, namely Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. This framework is integrated into a pipeline which makes the real-time system possible. Although there was no real method or benchmark to validate the results of the experiment, it is clear that this multimodal approach is feasible to a reasonable degree.

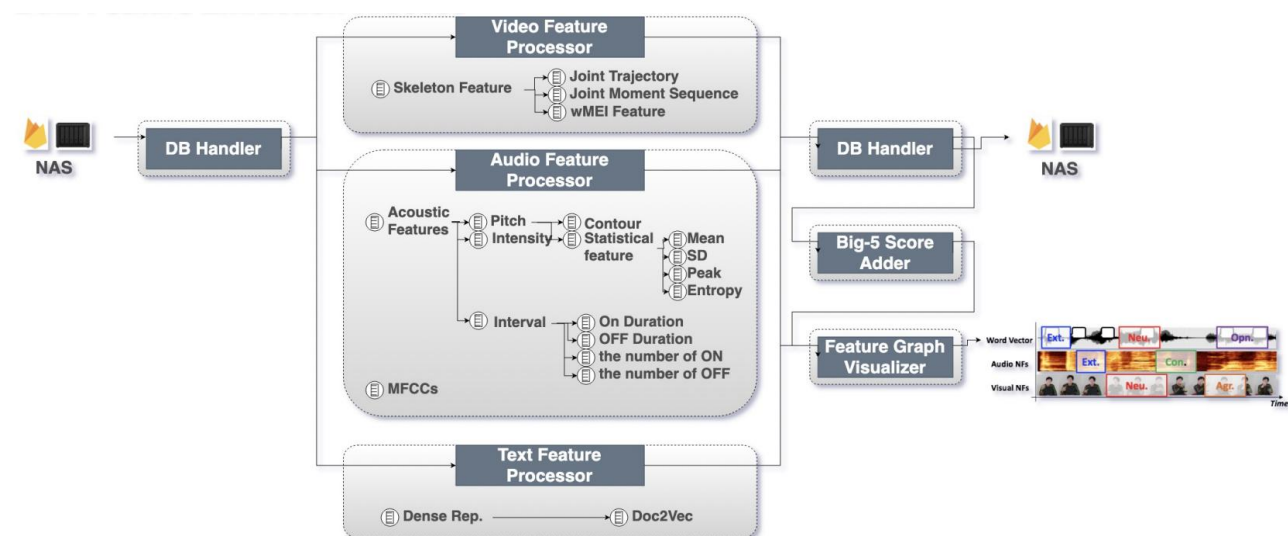


Fig. 3.11.1 Feature Extraction Process

3.12 Research Paper – 12

3.12.1 Paper Title

Zachary Yang, Nicolas Grenon-Godbout, and Reihaneh Rabbany. 2023, “Towards Detecting Contextual Real-Time Toxicity for In-Game Chat,” [12]

3.12.2 Summary

The model ToxBuster was introduced by the authors. ToxBuster is a real-time toxicity detection model focused mainly on the challenges seen in gaming platforms. An ever-growing platform with rapid messages being exchanged, and slangs related to different games, which needs moderation to try and control the harmful interactions promptly. It is visible that context plays an important role and decisions cannot be made by analysing individual messages. In order to resolve these complexities, the ToxBuster system uses Chat Data to analyse the sequence of messages, which helps in understanding the context. The system then checks metadata to capture timestamps, events that took part in the game, and player roles to provide more context to the model. The system is then made efficient enough to analyse real-time messages in-game. ToxBuster was trained and evaluated on many games such as Rainbow Six Siege, For Honor, and DOTA 2. This model was able to flag 82.1% of reported players with a precision of 90%. It was also able to identify an additional 6% of unreported yet toxic players.

3.13 Research Paper – 13

3.13.1 Paper Title

Przemyslaw Lenkiewicz, Binyam Gebrekidan Gebre, Oliver Schreer, Stefano Masneri, Daniel Schneider, and Sebastian Tschöpel. 2012, “AVATech — automated annotation through audio and video analysis,” [13]

3.13.2 Summary

The authors try the creation of annotations, which is really hard and takes 50 to 100 times the length of the annotated media, using automated annotations by learning through audio and video analysis. The model has 2 main roles, i.e., decrease the time taken to perform the annotations and using automation to increase the uniformity of the annotations, this can help maintain consistency. For audio and video analysis, the authors use Audio Segmentation, Speech Detection, Speaker clustering (label audio according to the speaker), Vowel and pitch contour detection (graphically use pitch contours to find similar patterns), Global motion detection (distinguish b/w different types of video content using motion), Hands and head tracking (use of a region-growing algo on seed points to create regions approximated by an ellipse). This method has helped reduce the average time taken to annotate by a factor of 0.23 to 0.38 as compared to manual annotating. The next focus of this method is to completely detect and track the movement of hands in a video and differentiate b/w the left and right hand.

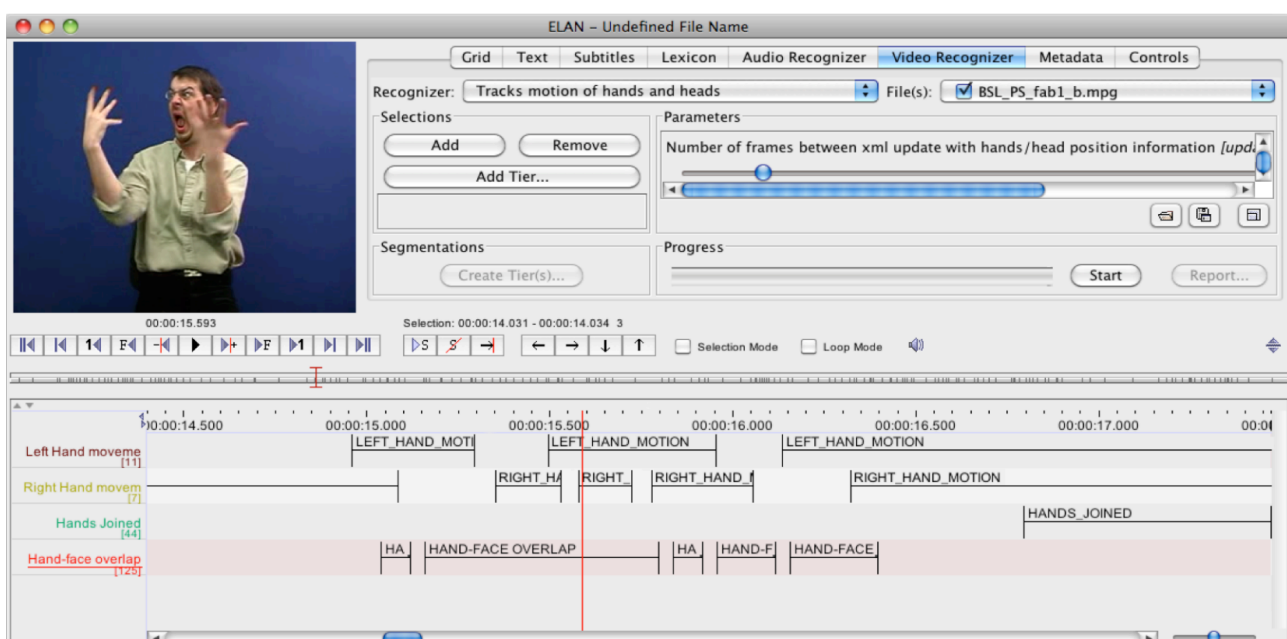


Fig. 3.13.1 Sample detection of motion for Annotator

3.14 Research Paper – 14

3.14.1 Paper Title

Jihyung Moon, Dong-Ho Lee, Hyundong Cho, Woojeong Jin, Chan Park, Minwoo Kim, Jonathan May, Jay Pujara, and Sungjoon Park. 2023, “Analyzing Norm Violations in Live-Stream Chat,” [14]

3.14.2 Summary

The goal of this paper is to find toxic language, such as hate speech, and norm violations in live streaming platforms like Twitch and YouTube Live. The study focuses on NLP to detect the aforementioned problem. The authors define norm violation categories in live stream chats and annotate 4,583 moderated comments (NormVio-RT is the dataset created by the authors) from Twitch. The model is trained based on informational context humans use in live stream moderation. The final results showed that using the right contextual information can increase the performance of moderation by 35%. The dataset contains moderated comments from the top 200 Twitch streamers, along with their chat rules. The main focus is on comments that triggered a moderation like user ban, or user timeout.

Coarse	Fine-grained	# Rules	# Violates		
			stage 1	stage 2	stage 3
Discrimination	Discrimination	13.98% (46)	2.34% (104)	2.25% (101)	2.34% (105)
HIB	HIB	22.49% (74)	21.33% (947)	26.55% (1,190)	27.80% (1,246)
Privacy	Doxing	0.60% (2)	0.34% (15)	0.36% (16)	0.36% (16)
Inappropriate Contents	Spoiler	0.60% (2)	0.02% (1)	0.02% (1)	0.02% (1)
	NSFW	1.82% (6)	0.86% (38)	0.85% (38)	0.85% (38)
	Self-destructive	1.21% (4)	0.32% (14)	0.29% (13)	0.29% (13)
	Illegal	0.30% (1)	0.16% (7)	0.07% (3)	0.07% (3)
Off Topic	Controversial Topic	5.47% (18)	0.59% (26)	0.85% (38)	0.83% (37)
	Begging	1.51% (5)	1.44% (64)	1.36% (61)	1.36% (61)
Spam	Excessive & Repetitive	11.24% (37)	17.59% (781)	21.64% (970)	21.42% (960)
	Advertisements	11.24% (37)	4.64% (206)	4.40% (197)	4.42% (198)
Meta-Rules (Live streaming specific)	Mentioning other streamers	14.28% (47)	0.72% (32)	10.62% (476)	10.58% (474)
	Backseating & Tall order	5.16% (17)	3.45% (153)	3.70% (166)	3.77% (169)
	Specific language only	10.03% (33)	0.97% (43)	6.94% (311)	6.94% (311)
Incivility (Miscellaneous)	Incivility	-	12.30% (546)	11.57% (519)	11.51% (516)
	Non-Identifiable	-	32.93% (1,462)	8.52% (382)	7.45% (334)
Total		329	4,439	4,482	4,482

Fig. 3.14.1 Statistics related to # of messages violating norms in different stages

3.15 Research Paper – 15

3.15.1 Paper Title

Wessel Stoop, Florian Kunneman, Antal van den Bosch, and Ben Miller. 2019, “Detecting harassment in real-time as conversations develop,” [15]

3.15.2 Summary

The authors find a method to detect video game players that harass teammates or opponents in chat. This would simplify the process for gaming companies to intervene during games by issuing warnings, muting, or banning the player. Toxic players are detected as the conversation develops, as early as possible, making it possible for gaming companies to intervene. The machine learning task is that instances (e.g.: players) change over time, as more information about the instances (more utterances) becomes available. This leads to time as an extra dimension of interest for metrics like precision, recall and F-score: instead of presenting them as a single number, it should be represented how they change during the conversation. The dataset consists of 5000 conversations from League of Legends, with utterances of 48,512 players. The proposed framework is called HaRe (Harassment Recognizer), it keeps track of toxicity estimates for all participants separately, updating the estimate for each utterance. This is done by concatenating all utterances, separated by [NEW UTTERANCE] tags, and classifying the resulting text. The rate in which the sliding threshold should be increased depends on the size of the training set: the larger the training set, the slower the threshold can be increased.

3.16 Research Paper – 16

3.16.1 Paper Title

Xiaoda Yang, Xize Cheng, Jiaqi Duan, Hongshun Qiu, Minjie Hong, Minghui Fang, Shengpeng Ji, Jialong Zuo, Zhiqing Hong, Zhimeng Zhang, and Tao Jin. 2024, “AudioVSR: Enhancing Video Speech Recognition with Audio Data,” [16]

3.16.2 Summary

The authors develop a VSR to predict spoken words by analysing lip movement in videos. To enhance VSR, they use audio data from a generative model for data inflation, including synthetic data with authentic visual data, and try leveraging the alignment of the audio and video. In zero-shot and full-shot situations, an audio-to-lip model was used for data inflation, using the knowledge learned from the generative model to enhance the lip-reading model. For cross-linguistic scenarios, the impact of different language families was considered on lip-reading tasks and later used to train an audio-lip-alignment model using self-supervised learning. Using TFG for data inflation is implemented by using a sequence model, AV-HuBERT which highlights the importance of temporal correlations in video data for VSR tasks. Audio inherently carries temporal information, so by transferring audio knowledge to the corresponding video, the audio-to-lip model generates a video stream that contains the same temporal information. The model significantly improves the semantic understanding of video content by using the temporal information in audio to strengthen the flow amongst video frames. AudioVSR is a unified model that maps the audio and video of all languages into the same space. With the AudioVSR model, they input audio from entirely new datasets and fine-tune on downstream tasks, developing a model suitable for VSR tasks.

3.17 Research Paper – 17

3.17.1 Paper Title

Athanasios Lykartsis and Margarita Kotti. 2019, “Prediction of User Emotion and Dialogue Success Using Audio Spectrograms and Convolutional Neural Networks,” [17]

3.17.2 Summary

The paper focuses on predicting dialogue success and user satisfaction, as well as user emotion, using only audio spectrogram representations and CNNs. The approach allows for a fast-running basic-time system that can enhance spoken dialogue systems (SDS). The speech itself contains verbal cues and emotions that can indicate user satisfaction. The dataset used is Let’s Go V2, which has 5065 audio interactions, with 3 labels – user emotion, subjective success, objective success. The audio was segmented into 1-second and 2-second for testing and mel-spectrograms were used. The best performing model was bottleneck CNN with 90% accuracy and the 1s spectrograms performed better than 2s.

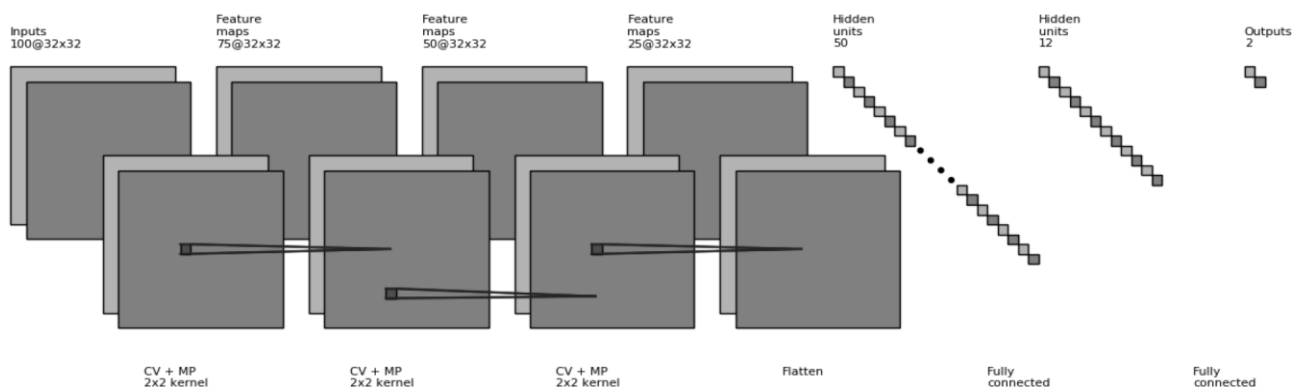


Fig. 3.17.1 Bottleneck architecture flowchart

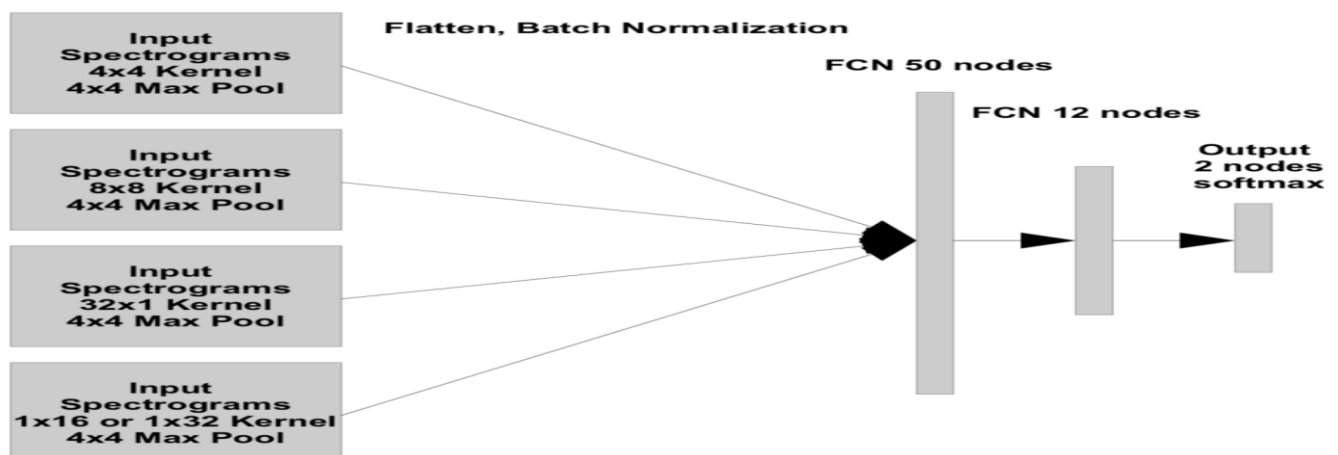


Fig. 3.17.2 Parallel architecture flowchart

CHAPTER 4

PROJECT REQUIREMENTS SPECIFICATION

4.1 Functional Requirements

- **Data Ingestion:** The system will take as input and extract video data in .zip files.
- **Audio Processing:** The system will transcribe audio using Whisper, detect profanity against a lexicon with the help of fuzzy matching, and calculate 30-dimension MFCCs with a 10ms hop rate.
- **Video Processing:** This system will be able to find a single face and extract a 50-dimensional geometric feature vector and interpolate the features to fit the 100 FPS audio timeline using MediaPipe.
- **Generation of Predictive Labels:** The system will produce predictive labels by shifting the profanity events, shifting by a 0.5-second horizon, making the model predict the future events.
- **Data Sequencing:** The system will segment the features into 1-second overlapping sequences and perform audio augmentation to positive (profane) samples.
- **Model Training:** The system will be trained to use late fusion on a multimodal CNNLSTM model. This is done by training an adaptive fusion layer to utilize the output of a pre-trained audio model, and use it to combine its predictions with those of a visual model.
- **Inference:** The system will take unseen video clips, and extract its features. This will be passed into the model. It will then use a median filter to smooth out predictions.
- **Censorship:** The system will recognize all the areas where the prediction probability is above a pre-established threshold, create a 1000 Hz beep tone and substitute the original audio at those areas.
- **Deliverable:** The system will result in a final, censored.mp4 video file that will be a combination of the original video stream and the altered audio track.

4.2 Non-Functional Requirements

- **Performance:** The inference pipeline should be optimized to have low latency. It is done through downsampling video input (15 FPS processing) and employing a lightweight visual model.
- **Reliability:** The system should be resistant to video quality problems. The adaptive fusion layer will be made to automatically switch to the "audio-only" prediction when the face prediction confidence is low or when there is no face.
- **Accuracy:** The main success measure of the system will be F1 score. The probability limit should be optimized in order to maximize this score.
- **Usability:** It will have clear console feedback and progress bars on all long-running processes.
- **Maintainability:** All the important parameters (e.g. SAMPLERATE, FUZZYTHRESHOLD, WINDOW_SIZE) will be represented as global constants so that they can be easily configured.

CHAPTER 5

SYSTEM DESIGN

5.1 System Architecture

The structure is made to be a three-stage modular pipeline, a back-end process whereby the output of one stage will be the input of the other:

- **Preprocessing Pipeline:** Accepts raw video information and converts it to training data in the form of labels and numbers.
- **Training Pipeline:** Uses the processed information in order to construct, train, and store the final multimodal model.
- **Reasoning Pipeline:** This final model is used on a new unseen clip to create an output file which is censored.

5.2 Component Design

The system is subdivided into a number of important modules with each having a particular responsibility:

- **Audio Module:** It employs the OpenAI Whisper to perform the transcription and fuzzywuzzy to find profanity. It then uses librosa to compute 30-dimension MFCC features at hop rate of 10ms.
- **Video Module:** It is based on MediaPipe, which extracts 50 geometric face features (e.g., MAR, EAR) on a single face that is identified. It interpolates between the 15 FPS video features with the 100 FPS audio timeline using the numpy interpolation.
- **Data Augmenter & Sequencer:** Slices full-length features into 1.0 -second, overlapping windows. It then time stretches (audio augmentation), pitches the positive (swear) samples, and adds noise to balance out the dataset to train with.
- **Training Module:** This module was developed in PyTorch and it specifies the CNN-LSTM model structure and trains the system using a two-step transfer learning method (audio-only, followed by fusion).
- **Inference & censorship module:** This is a self-executable script that loads the trained multimodal model, and uses the entire preprocessing and inference pipeline on a new video. It then replaces any profanity with a 1000 Hz beep using MoviePy and pydub.

5.3 Key Design Choices

The effectiveness and reliability of the system is determined through three critical design choices including:

- **Predictive Labelling:** This system will be predictive and not only reactive. By delaying the labels of the training by 0.5 seconds, the model has to be trained to predict a profanity occurrence. This enables the bleep to be applied when the event is occurring and not when it has already taken place.
- **Hybrid CNN-LSTM Model:** The model was selected because it uses the advantages of both models. The 1D-Convolutional (CNN) layers are good in the process of detecting the local, time scalar patterns in the MFCCs and face features whereas the Long Short-Term Memory (LSTM) layers do the temporal order of the patterns.
- **Adaptive Late Fusion:** This is the essence of the multimodal design. The system not just concatenates features, but rather, it trains both audio and visual models and at the end combines their predictions. AdaptiveFusionLayer trains to perform the weighting of these predictions and is specifically trained to drop to the trusted audio-only prediction when face confidence is low (e.g. no face detected, bad lighting). This option directly meets the non-functional requirement of reliability and robustness of the system.

CHAPTER 6

PROPOSED METHODOLOGY

6.1 Data Preparation

The data preparation workflow is a process that acquires raw video data, extracts and preprocesses the audio signals, matches it to a time-aligned transcript, and finds the instances which contain profanity. Initially the video to be processed is downloaded from sources such as YouTube, Twitch etc. The scraper ensures that the configuration is set to the best video and audio streams, before creating a standard MP4 file. From this MP4 file, the audio track is converted into a compatible format which is used for transcription by the model. To ensure that the performance does not hinder, the waveform is segmented into 5-minute chunks, which are exported to temporary WAV files, and then transcribed individually. A word search is performed on these transcriptions to find the timestamps of the predefined profane words. Using these instances, we generate video clips that make the dataset. The clips in this dataset are then processed to ensure that they are not liable to pronunciation errors, transcription faults, and variations in target words. The clips are further analysed at the frame-level, where frames containing profanity are labelled as 1 and the remaining as 0. This establishes an alignment between the frames and the audio features for each clip, that is factually correct. In order to build real time systems, we need to focus on prediction rather than classification. This is done using a window to shift the labels, such that profane instances occur after 0.5 seconds of the label. A summary csv file is created to store the metadata of the processed clips.

6.2 Feature Extraction

We extracted two separate feature streams in order to record both visual and auditory data. For the audio modality, the audio track of each video was resampled to 16kHz. We then extracted 30-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) using a 10ms hop length. This resulted in a feature matrix of shape $(30 \times T)$, where T is the number of 10ms frames. In the case of visual modality, we have applied the MediaPipe FaceMesh framework to identify and track 478 facial landmarks. Rather than operating with the raw landmarks, we used temporal spatial correlations to compute a 50-dimensional geometric feature vector, which captures dynamic spatial relationships such as Mouth Aspect Ratio (MAR), Eye Aspect Ratio (EAR), lip distances, jaw angles etc. Video

frames were handled at a lower rate of 15 FPS to optimize the performance to real time. This sparse visual feature stream was then temporally interpolated to match the 100 FPS (10ms hop) resolution of the audio features. This resulted in a final, synchronized feature matrix of shape $(50 \times T)$, where T is the same number of 10ms frames as the audio.

6.3 Data Sequencing and Augmentation

In order to prepare the data to be used in training, we initially chopped the long stream of features into sequences that were 1.0 second long with a stride of 0.2 seconds, to generate overlapping samples. ``Fig. 6.3.1'', shows that the most difficult part was that the data is highly skewed, i.e., the majority of clips were clean, and it was extremely rare to find the samples containing swear words. We discovered that normal balancing methods have actually worsened the model and made it over predict profanity on normal speech. This was solved using a multi-part approach:

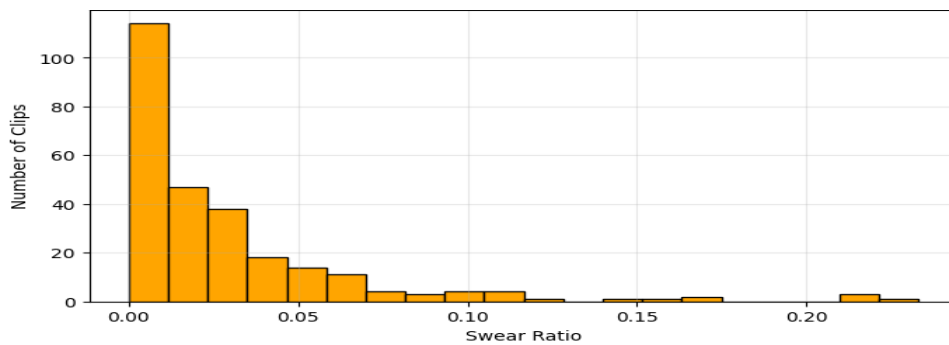


Fig. 6.3.1 Swear Frame Ratio per clip

- **Predictive Labeling:** Each of the sequences was assigned a predictive label. We had a view of 0.5 seconds ahead of the window, and in case a swear word appeared in the future segment that we looked at, we scored the whole 1.0 second window as positive.
- **Selective Augmentation:** To obtain more of these rare samples of positive data, we applied audio augmentation, i.e., time-stretching, pitch-shifting and adding noise only to the swearing clips. We then made copies of the original face features in order to make matching pairs while maintaining the class balance.
- **Weighted Balancing:** We retained all our new positive sequences and reduced the amount of the non-profane class to a smaller ratio, while preserving the natural class distribution by maintaining a higher proportion of the non-profane class, so that our model learns a true prior instead of an artificially imbalanced dataset. We also trained the model with a weighted loss function to make the model learn on the remaining rare profane samples. This trains the model to be much more careful about profane samples and increase the penalty in case it incorrectly classifies a "swear" sample.

6.4 Audio Module

The audio module is the most important pillar of the model. The audio model is the primary and most robust component that predicts an upcoming swear word. The audio module balances both high predictive capabilities and low latency in its prediction speed for its usage in real-time scenarios. This component can be broken down into two distinct aspects which are the acoustic feature representation as well as the temporal modelling and the pattern recognition done by the hybrid deep learning architecture.

1. **Acoustic Feature Representation:** Instead of raw audio waveforms we decided to use Mel-Frequency Cepstral Coefficients (MFCC) as it imitates the non-linear frequency perception of the human auditory system. Thus, it helps the model understand how the human being is perceiving the audio and can learn through that perspective. The multi-step process used in converting the audio to MFCCs involves:

- **Resampling and Normalization:** All the incoming audio is resampled to a uniform 16000 Hz which is a widely accepted rate for ensuring consistency in the data.
- **Framing and Windowing:** The signal is then converted into windows of 25ms with an overlap of 10ms. These continuous windows are an indispensable tool which allows us to feed the model the exact starts of words.
- **Frequency Domain Transformation:** To get a magnitude spectrum, a Fast Fourier Transform (FFT) is done on the signal to convert it from the time domain to a frequency domain.
- **Mel Filterbank Application:** By applying a bank of triangular filters, the power scale is fit on to the Mel scale. Each filter's frequency band's energy is added.
- **Logarithmic Compression:** We then take a logarithm of the filterbank as the human perception of loudness is logarithmic as well.
- **Discrete Cosine Transform (DCT):** A Discrete Cosine Transform is performed on the log filterbank energies which gives us our MFCCs. Another thing of note is the energy compaction property of DCTs which result in the lower-order coefficients containing the most vital data.
- **Final Parameterization:** The process results in a 30-dimension MFCC vector for each 10ms hop. To yield a rich representation without too much complexity, this dimensionality was chosen. For a 1 second clip, we get a final output of a tensor of 30 cepstral coefficients over 100-time steps.

2. Hybrid Deep Learning Architecture: The core architecture contains a Convolutional Neural Network (CNN) for translation-invariant detection and a Long Short-Term Memory for modelling long range temporal dependencies. This helps the model learn the shape as well as sequence of shapes that the signal is producing.

The CNN Architecture,

- **First CNN Block:** This block processes the 30-dimensional MFCC input with 64 convolutional filters. A kernel size of 3 allows it to capture patterns across three adjacent time steps, while a padding of 1 ensures that the temporal dimension is preserved.
- **Second CNN Block:** The second block takes the 64-channel output of the first block and applies 128 filters to it, thus learning the higher-level feature combinations.
- **Stabilization and Regularization:**
 - **BatchNorm1d (Batch Normalization):** Batch Normalization normalizes the activations of the previous layer which combats the problem of internal covariate shifts. This stabilizes the training process, allowing for higher learning rates and provides a slight regularization effect.
 - **ReLU (Rectified Linear Unit):** ReLU is a non-linear activation function that allows the network to learn complex and non-linear relationships. It is non-saturating in nature and helps mitigate the vanishing gradient problem.
 - **Dropout:** It is a technique that randomly sets a selection of the neuron activations to zero during training. By being unable to be overly reliant on any one feature, we force our model to generalize on unseen data. Our model uses a dropout rate of 0.2.

The LSTM Architecture,

- **Stacked LSTM:** A two layer Long Short-Term Memory model was used for hierarchical learning. For example, the first layer may learn acoustic patterns to syllables and the second layer may learn syllables to words or phrases.
- **Hidden State Dimensionality:** Each LSTM layer has a hidden size 128, which ought to be large enough to represent all the complicated temporal dynamics which occur in a one-second audio file.
- **Final Context Vector:** The model obtains the final hidden state of the last time-step of the second LSTM-layer which is a single compressed-vector which represents the entire one second audio sequence.

3. The Final Classifier: To project the 128-dimensional output to 64-dimensional space, the final classifier contains a linear layer which is followed by another ReLU activation and a Dropout of 0.3. The training methodology and systemic role involves the following:

- **Loss Function and Class Imbalance Mitigation:** The BCEWithLogitsLoss function is used to optimise the model and a key parameter is the pos_weight which is calculated as the ratio of negative samples to positive samples in the training set. This weight increases the loss incurred when the model misclassifies a positive sample which allows us to fight the bias the model may have towards the majority class and thus improving the model's performance.
- **Optimization:** Adam is an adaptive learning rate optimization algorithm that has the advantages of both AdaGrad and RMSProp which is used to update our model's weights.

6.5 Video Module

The Video module is a component that helps the primary audio module to better its prediction by providing it with crucial facial data such as mouth movements, eye movements, emotional expressions, etc. The focus on audio instead of video is to reduce the latency of the model working in real time.

Our system uses a complicated feature engineering method to get around the high cost of processing raw video pixels with deep 2D or 3D CNNs. Using the MediaPipe framework from Google, we turn the video frames into a low-dimensional but high-information geometric feature vector.

This results in dimensionality reduction as well. Raw video frames are extremely high-dimensional. Processing such data directly in real-time is expensive and time-consuming and by comparison, MediaPipe's abstracts each frame into a set of 478 3D landmarks, from which a mere 50-dimensional feature vector is engineered which represents a massive reduction in data complexity. This represents a massive reduction in data complexity. Geometric data is also much more invariant than raw video pixel data.

The feature generation pipeline involves:

- **Temporal Down sampling:** We only process every other frame in a 30 FPS video, meaning a target rate of 15 frames per second, that halves the computational load which is very important for a model that requires very low latency.

- MediaPipe for Landmark Detection: Each frame that was selected is run through the face_mesh MediaPipe model that identifies a single face and produces a mesh of 478 3-D features that correlate to important parts of the face such as lips, eyes, eyebrows, nose, and jawline.
- Geometric Feature Engineering: The landmark coordinates are used to generate a 50-dimensional feature vector. These capture,
 - Mouth: A majority of the features we try to collect are focused on the mouth such as the Mouth Aspect Ratio, which is the ratio of the vertical opening of the mouth to its horizontal opening.
 - Eye: We can use the widening or squinting of the eyes as indicators of emotions and we calculate the Eye Aspect Ratio for both the eyes.
 - Eyebrows and Jaw: The eyebrows being raised or the movement of the jaw can give us important features which can help us identify when a profanity is going to be uttered.
 - Global and Relational Features: The Face Aspect Ratio is an example of a global feature along with other relationships collected.
 - Temporal Synchronization: The feature vectors, which were originally produced at 15 FPS, are on a different time scale as compared to the 100 FPS audio MFCCs. To integrate these modalities, there has to be a complete alignment. Linear interpolation is used to achieve this. The 15 FPS sequence of visual features is up sampled to 100 FPS creating new feature vectors for the intermediate time steps. The weighted average of the values from the two nearest original frames is used as the value of the new feature. Every 10ms audio feature now has a corresponding time-aligned video feature.
- 1. Lightweight CNN-LSTM Architecture: The neural network architecture for the visual model is designed to be around the same as the audio model, employing a CNN-LSTM hybrid. However, it is intentionally designed to be significantly lighter and computationally cheaper.

The CNN Layer Architecture,

- Functionality: Similar to the audio model, the Conv1d layers detect local, time-invariant patterns in the sequence of geometric features. Here, they learn to recognize short, characteristic facial movements.
- Architectural Differences: The visual CNN is shallower and narrower than its audio counterpart.
 - First CNN Block: Processes the 50-dimensional input with only 32 filters.

- **Second CNN Block:** Expands the representation to 64 feature maps. This is a marked reduction compared to the audio model's 64 to 128 filter structure, leading to a substantial decrease in the number of parameters.

The LSTM Layer Architecture,

- **Functionality:** To process the sequence of feature maps to model the change in facial gestures over time, we use an LSTM model.
 - **Architectural Differences:** Instead of using a 2-layer architecture, like in the audio model, we use a model with only a single layer.
2. The Final Classifier, i.e., fully connected layers includes a classifier head that is also streamlined, with a smaller number of neurons in its hidden layer, further contributing to the model's overall efficiency.

6.6 Inference and Censoring Pipeline

In order to implement the trained model, we constructed an inference pipeline that processes videos that have never been seen. "Fig. 6.6.1" shows that the pipeline initially extracts and synchronizes audio (MFCC) and facial (geometric) features, as done in training. It then processes these features using the model with the same 1.0-second sliding window in order to produce a continuous probability stream.

We use a temporal median filter to smooth these predictions so as to reduce false positives. The smoothed probability is compared against the threshold, and a "bleep" is scheduled into the system when the probability is greater than the threshold value. Since the model was trained with a 0.5-second prediction horizon, the model is capable of muting the audio before the swear word is said. Lastly, the system cuts the original video and rejoins it with this now bleeped audio track to produce the final censored video.

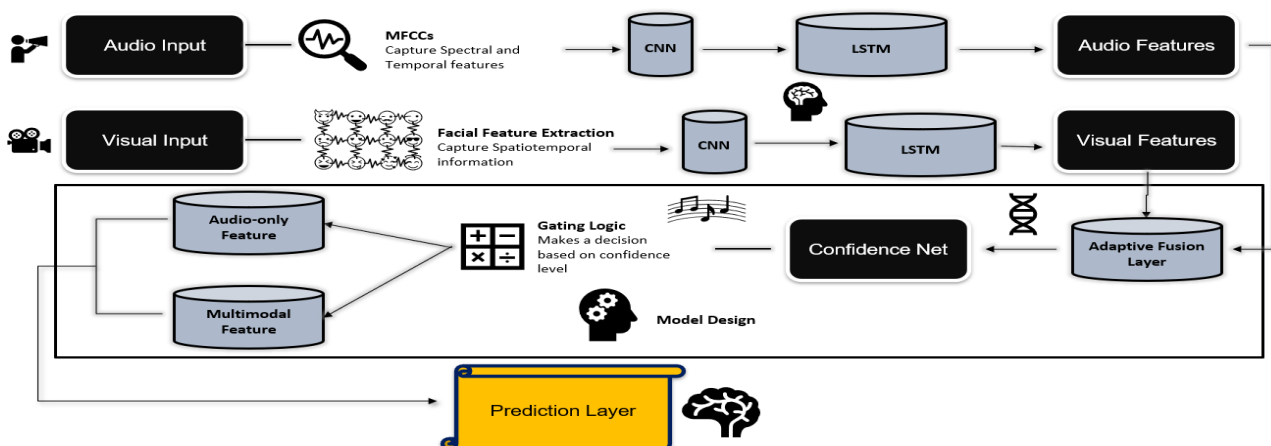


Fig. 6.6.1 Model Architecture Design

CHAPTER 7

RESULTS AND DISCUSSION

This section shows the quantitative and qualitative findings of the model training and assessment. The experiments were aimed at determining the high-performance audio-only state, testing the functioning of the final multimodal system, and determining the possibility of using the system in real-time.

7.1 Experimental Setup

The augmented and sequenced data were 58,404 sequences, each one second long. The data set was divided into training (46,723 samples), validation (5,840 samples) and test (5,841 samples) sets, with the stratified positive class (profanity) ratio of 16.7%.

All of the models have been trained on a CUDA-enabled GPU. In order to cope with the large amount of class imbalance, `pos_weight` of 5.0 was added to the `BCEWithLogitsLoss` function to add the penalty on the misclassifications of the rare positive class.

7.2 Audio-only Model Performance

An audio-only CNN-LSTM baseline model was trained to create a performance baseline. The model showed high discriminative ability as it had a Test AUC of 0.959. Once the decision threshold on the validation set was optimized to 0.873, an increased result over the default of 0.5, the model reached a final Test F1-score of 0.826. The model had a high recall of 0.823 on the profanity class.

7.3 Multimodal Model Performance

One of the main peculiarities of the dataset was that only 7.0 per cent of the training sequences included any facial features that could be detected, which restricted the data supplied to the visual branch.

The final multimodal model had a Test AUC of 0.960 and Test F1-score of 0.824, at optimal threshold of 0.826 which can be seen in ``Fig. 7.3.1 ". The results of this model yielded a statistically significant

improvement in recall of profanity to 0.859, which identified a greater number of true positive cases than the baseline, although with a little less precision.

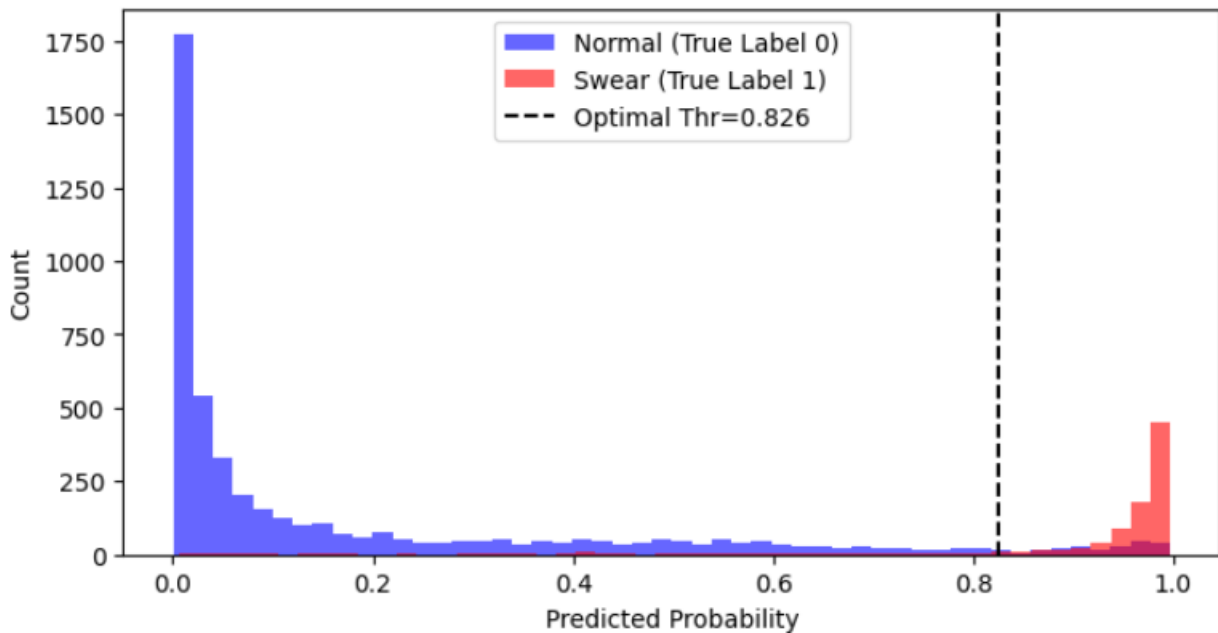


Fig. 7.3.1 Prediction Probability Distribution by True Class

7.4 Comparative Analysis

When the two models are compared directly, it is seen that final F1-scores were close to each other. The behavior of the multimodal model however changed to favor higher recall.

The trained AdaptiveFusionLayer was analyzed and it was found that it had learned to give a weight of 0.90 to the audio stream and a weight of 0.06 to the visual stream. This means that the fusion model correctly identified that the audio stream is to be used as the main and most accurate predictor due to the lack of visual data. Effectively, the adaptive model was able to learn how to revert to a superb audio-centric model and use the available limited visual data in order to improve recall.

7.5 Beyond the Audio: Facial Cues to increase the accuracy of profanity recognition

While testing on a clip extracted from a stream, consisting of 2 swear words - one at 3 secs and the other at around 6 secs, the audio specific model predicts numerous profane occurrences as shown in

Fig. 7.5.1", while the final multimodal model correctly eliminates false positives as shown in Fig. 7.5.2".

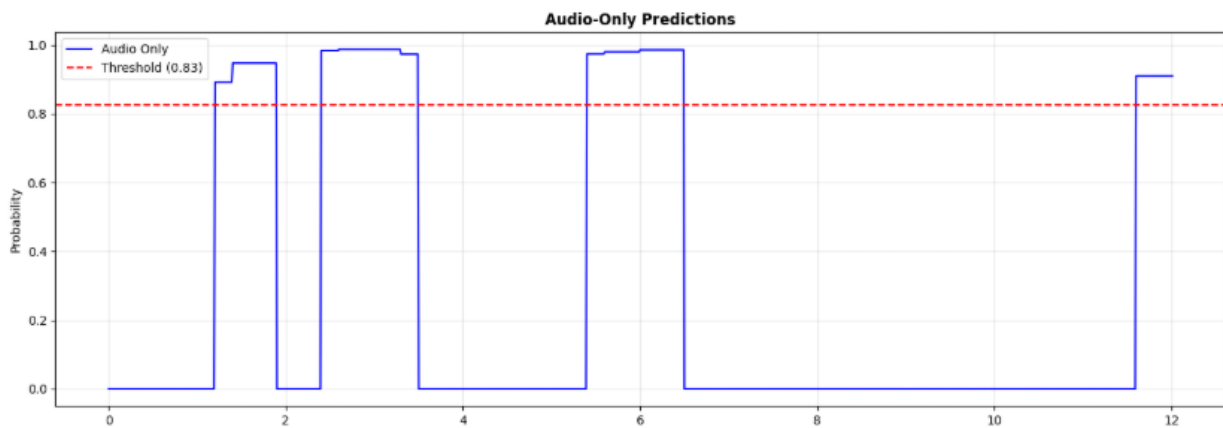


Fig. 7.5.1 Audio-Only Prediction on a live sample

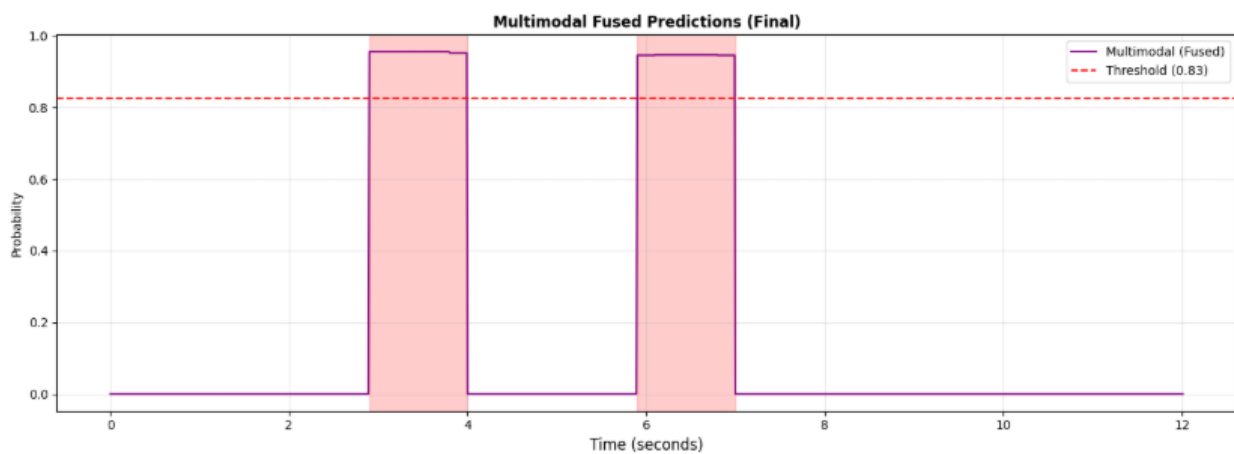


Fig. 7.5.2 Final Multimodal Prediction on a live sample

7.6 Latency and Real-Time Feasibility

A latency test was carried out to test the validity of the system to be used in real-time. The complete multimodal model proved to be very efficient with an achievement of:

- Average Latency per Sample: 0.036 ms
- Throughput: 27,688 samples/second

These findings ensure that the computational cost of the model is insignificant and has the potential of executing in a real-time, or faster-than-real-time, processing environment.

CHAPTER 8

CONCLUSION AND FUTURE WORK

In this paper, we explored the notion of real-time profanity prediction in live streams and demonstrated how a multimodal model (Audio and Video) can achieve this feat with high accuracy and low latency. Through systematic analysis, we showed that our system can process audio and video streams and is suitable for real-time deployment in practical scenarios.

Overall, our work highlights the potential of a lightweight architecture for real-time profanity detection.

Our model can be improved to consider a higher weightage for facial features, lower false positives and improve latency. This can be achieved by training on a larger amount of data. Latency is real-time but currently in the context of a GPU environment, this can be reworked to function in any environment regardless of hardware.

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